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ISYE 7406: Data Mining & Statistical Learning

Project Final Report

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“Why is the water so murky in the lakes near Atlanta?”

Abstract

The goal of this study was to identify the key factors causing murky and eutrophic lakes in the continental U.S. Based on data from the U.S. EPA National Lakes Assessment, I used Random Forest and piecewise SEM models to determine that phytoplankton concentration is the main cause of lakewater turbidity. I also found that bottom-up forces (specifically nitrogen and phosphorus concentrations) are the most important factors in phytoplankton growth, which had a greater magnitude and more consistent effect than top-down ecological forces. Some interesting ecological interactions were still detected by the models, however, and the populations of zooplankton and benthic macroinvertebrates do seem to have some influence on both phytoplankton and macroalgae. Lastly, macroalgae and phytoplankton seem to engage in competition for phosphorus, with phytoplankton concentration increasing with phosphorus and the inverse for macroalgae. Overall, it seems like the best way to improve lake habitat is to reduce the input of excess nutrients.

Introduction

As a resident of metro-Atlanta who enjoys outdoor activities, I frequently find myself on a warm summer day excited to spend some time at the nearby lakes Allatoona or Lanier, only to arrive and be immediately grossed out by the appearance of the water. It tends to be murky with a green hue, dead fish odor, littered with debris, and the type of muddy that sucks up your feet or sandals to the point where you sometimes can barely walk. Not to mention the rumors of brain-eating amoebas lurking in the depths… These experiences make me wonder whether this is a common experience around the U.S. or if it’s unique to these man-made lakes of the Southeast. I have a background in ecology and even worked with freshwater algae for my thesis, so naturally I am quite interested in pinpointing the possible causes of these undesirable recreational conditions, as well as any potential related consequences to the ecosystem. For this project I aim to analyze the factors that contribute to the murky conditions of U.S. lakes and propose potential solutions if possible.

My analysis is based on the 2022 National Lakes Assessment survey by the EPA (more information in the Data Sources section). They found that 49% of U.S. lakes were high in chlorophyll-a (a proxy for phytoplankton growth), 47% are high in nitrogen, and 50% are high in phosphorus. All of these percentages have been increasing since 2012. They also found a number of additional stressors to healthy lake ecosystems, including acidification (low prevalence but high impact), herbicide presence, lakeshore disturbance, and limited riparian and littoral habitat. Nitrogen and phosphorus are key nutrients that lead to phytoplankton growth and eutrophication. 30% of U.S. lakes are currently classified as eutrophic, as opposed to 26% combined oligotrophic or mesotrophic which are typically considered healthier (for recreation and harmful algal bloom potential). Increasing nutrient concentrations are typically attributed to anthropogenic sources like agricultural runoff of fertilizer and manure, sewage overflow (which I believe is common in Atlanta after large storms), and runoff from golf courses and lawns.

Georgia is also known for its red clay and forests, which could both reasonably cause murky water in the Atlanta area. Leaf litter from trees leach tannins into water and cause brown coloration and potentially slight pH differences, although I believe leaf litter is often considered healthy overall for lake ecosystems. Phytoplankton growth could also be controlled by healthy herbivore populations and the rest of the aquatic food web including zooplankton, benthic macroinvertebrates, fish, and birds. I'm interested in how much of lake murkiness can be attributed to clay, tannins, or phytoplankton, and whether phytoplankton can be limited by healthy food webs or solely via nutrient reduction.

Problem Statement

Lakewater visibility has widespread recreational and ecological implications. The goal of this analysis is to identify the relative contributions of key factors that impact U.S. lake ecosystems so the most important influences can be identified and addressed. Specifically, I want to determine whether nutrient availability or ecological food webs have a stronger impact on phytoplankton concentration. Bottom-up control via nutrient availability would signify that anthropogenic factors like fertilizer application and runoff should be the main focus, whereas top-down control via herbivory and a healthy food web would indicate that habitat restoration and species protection is more important (although there are certainly overlapping anthropogenic factors on both sides).

Data Sources

This analysis combines the Environmental Protection Agency's "National Lakes Assessment" (NLA) dataset (US EPA, 2024) with iNaturalist observations (iNaturalist) accessed via GBIF.org. The NLA dataset contains 4 years of data collected from ~1000 U.S. lakes (981 sites in the 2022 survey), representing a stratified random sample of 268,020 lakes meeting their search criteria. The dataset contains many useful features, including counts of benthic macroinvertebrates, zooplankton, and phytoplankton; lake surface area and surrounding habitat; water depth and visibility; and concentrations of key nutrients and other important biomarkers including total nitrogen, phosphorus, and chlorophyll-a. This dataset also contains summary statistics from surrounding watersheds derived from the LakeCat dataset (Hill et al., 2018), including the amount of farmland, human development, forested area, and much more. I used only the 2022 data for this analysis.

I merged the NLA data with iNaturalist observations of certain groups of birds seen near each lake around the same time period (Jan-Dec 2022). I created new variables based on the number of species observed within 4 taxa of birds typically associated with aquatic habitats: Anseriformes (Ducks, Geese, Swans), Coraciiformes (Kingfishers, etc.), Pelecaniformes (Pelicans, Herons, Cormorants), and Accipitriformes (Hawks, Eagles, Osprey). I matched an observation with a lake if the observation location is within a mile of the lake's latitude and longitude coordinates. Only observations tagged as "Research Grade" were considered.

Methodology

Data Preparation

Feature Engineering

In order to aid model convergence and normality assumptions I log-transformed all count, concentration, and some percentage data (see Appendix 1 for details). I also created the clay percentage feature based on silt and sand measurements, since soil is composed of silt, sand, and clay particles.

For the bird species counts, I merged bird observations with each lake by calculating the Euclidean distance in miles between the coordinates of each bird sighting and the lakes using the LAT_DAT83 and LON_DD83 features. I scaled latitude and longitude distances by a factor of 69 to convert degrees to miles and also scaled longitude by cosine(latitude). I kept all observations within 1-square mile of the lake coordinates, then aggregated by species across the 4 focal taxa.

Missing Data

The initial dataset contained around 154 observations with missing values in many features. I'm not sure why these lakes were missing so much data so I removed them. The remaining features contained only 7 or fewer NAs each, so I simply omitted these rows. The resulting dataset contained 950 unique lake observations, all between May and September 2022.

Multicollinearity

I checked the dataset for highly correlated features (Appendix 2) and evidence of multicollinearity. There were strong correlations between nitrogen, phosphorus, chlorophyll-a, and turbidity, which are all explained by interactions between phytoplankton, nutrients, and turbidity and are accounted for in the model architecture. There were also strong correlations between all 4 bird taxa, likely because the majority of all values were zero and other biases in the dataset.

I also checked for multicollinearity using variance inflation factors and all scores were less than 5, which is a common threshold suggesting covariance between features should not be an issue.

Model Selection

Model Architecture

To determine the most influential factors on lakewater turbidity and phytoplankton concentration, I created a hierarchical interaction graph based on prior knowledge. First, a 3-feature model of turbidity (a measure of water clarity), followed by identical 15-feature models of chlorophyll-a (a proxy for phytoplankton concentration) and an index of macroalgae presence. All features and model architectures can be seen in Figure 2.

To model these hierarchical interactions, I fit three Random Forest models (one for each response variable) and a piecewise Structural Equation Model (SEM), which is composed of 3 linear models where the features are arranged as a directed acyclic graph (DAG). I also included residual variance between macroalgae and chlorophyll-a in the piecewise SEM model to account for mutual competition for resources between macroalgae and phytoplankton.

These models both have distinct strengths and weaknesses for this analysis. Random Forest models do not make assumptions on the data distribution and are not impacted by skewed data. They can also easily fit nonlinear relationships. The downside is that they are more difficult to interpret and compare, so I used partial dependence plots (PDPs) to visualize the marginal influence of each feature on model output. Piecewise SEM models compute easy-to-interpret linear coefficients for each feature and the coefficients are multiplicative along paths in the DAG. However, they cannot model nonlinear interactions and assume the residuals will be normally distributed and homoscedastic. I did not optimize any hyperparameters for either model due to computational constraints.

Statistical Testing

To compare Random Forest and piecewise SEM model performance and feature importance, I ran 100 rounds of Monte Carlo cross validation with 80/20 train-test split. During each round, I resampled the training and test sets before standardizing the data based on the column means and standard deviations of the training set to avoid leakage. I predetermined the prediction grid for PDP calculations based on the column min and max values of the full dataset before sampling so the PDPs would be within the appropriate range and able to be aggregated across rounds. This does not have any influence on the models and thus does not cause leakage. During each round of Monte Carlo CV, I trained all three Random Forest models and the piecewise SEM on the training data, then calculated R-squared on the test data, calculated the PDPs, and collected the feature importances and coefficients from each model. The models all used the same training samples so their performance could be compared using paired t-tests. When aggregating the results I calculated both 95% confidence intervals and 95% quantile ranges, and sometimes I chose to display one or the other in the figures below depending on which I felt was more appropriate.

Results

Model Comparison

The piecewise SEM and Random Forest models performed similarly based on R-squared values calculated on the test sets across 100 rounds of Monte Carlo CV (Figure 1). There was no statistical difference between the R-squared values for log(Turbidity) or log(Chlorophyll-a), but Random Forest performed better for the Macroalgae Index response variable (paired two-sided t-test, $t = 20.141$, p-value < 0.001). The models for log(Turbidity) and log(Chlorophyll-a) seem to have reasonable predictive power, with R-squared values greater than 0.5. However, the Macroalgae Index models – with R-squared less than 0.25 - might not capture enough variance to draw many ecological conclusions.

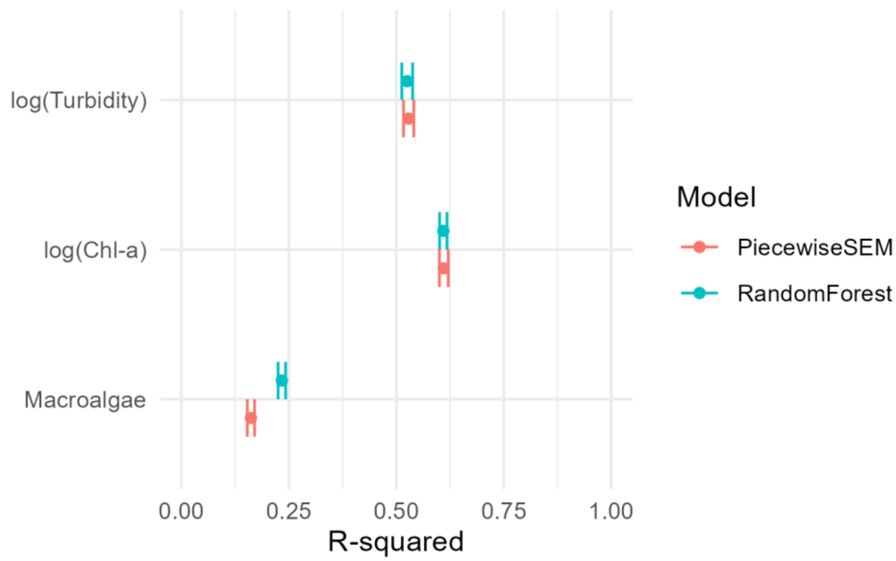
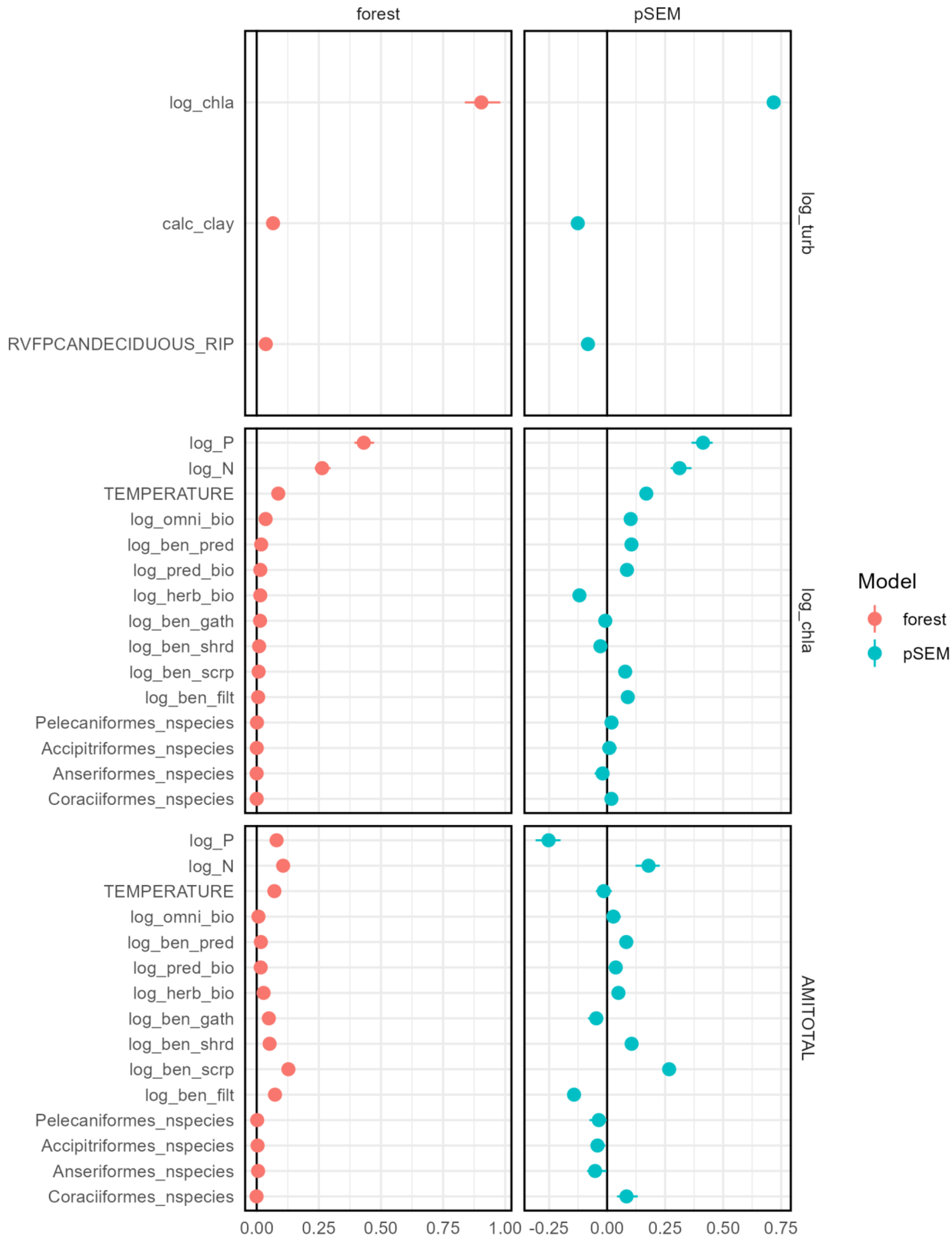


Figure 1: Piecewise SEM and Random Forest model performance. R-squared was calculated on the test set of each Monte Carlo CV round, and the mean +/- 95% CI is displayed for each response variable.

Feature Importance

Feature importance for Random Forest (RF) models and coefficients for piecewise SEM (pSEM) models are displayed in Figure 2. Chlorophyll-a concentration was by far the most important feature explaining turbidity in both RF and pSEM models. Phosphorus and nitrogen concentrations had the greatest influence on chlorophyll-a concentration, while water temperature and various ecological factors had a marginal effect. The number of bird species present in key taxa had no influence on chlorophyll-a concentration in either model. For the Macroalgae models, phosphorus and nitrogen once again seem to be key factors, but the populations of benthic scrapers and filterers also seem influential. Interestingly, high phosphorus concentration seems to reduce macroalgae presence while the opposite is true for nitrogen. Benthic scrapers and shredders seem to marginally increase macroalgae presence while filterers may have a negative influence. Once again, the influence of bird taxa is minimal, with only a small positive coefficient for Coraciiformes (e.g. kingfishers) in the pSEM models.



forest
pSEM
log_chla
calc_clay
RVFPCANDECIDUOUS_RIP
log_turb
log_P
log_N
TEMPERATURE
log_omni_bio
log_ben_pred
log_pred_bio
log_herb_bio
log_ben_gath
log_ben_shrd
log_ben_scrp
log_ben_filt
Pelecaniformes_nspecies
Accipitriformes_nspecies
Anseriformes_nspecies
Coraciiformes_nspecies
log_chla
AMITOTAL
0.00 0.25 0.50 0.75 1.00
-0.25 0.00 0.25 0.50 0.75
Model
forest
pSEM

Figure 2: Feature importance (Random Forest) and model coefficients (piecewise SEM) for each response variable. For Random Forest models, feature importance represents the increase in MSE when each variable is permuted. For piecewise SEM, the model coefficients are displayed. The error bars represent the 95% quantile range across 100 rounds of Monte Carlo CV. For piecewise SEM, the residual covariance between log_chla (chlorophyll-a) and AMITOTAL (macroalgae index) was –0.054 with 95% quantile range (-0.086, -0.026).

Partial Dependence

Partial Dependence (PD) plots for Random Forest models illustrate the magnitude, direction, consistency, and linearity of the influence of each feature on overall model predictions. For the turbidity models, chlorophyll-a concentration had a strong, monotonically increasing influence that was highly linear near the center of its distribution and flattened out at the extremes (Figure 3a). Neither clay nor deciduous forest cover had a consistent impact. Phosphorus and nitrogen concentrations also had strong monotonically increasing relationships with chlorophyll-a concentration, once again saturating at extreme values (Figure 3b). Water temperature and omnivorous zooplankton biovolume also seem to have a marginal (yet inconsistent) positive effect, similar to the pSEM models. The PD plots for Macroalgae Index are highly variable, highlighting the lower predictive power for this response variable (Figure 3c). Macroalgae presence seems to increase with benthic scraper populations and decrease with phosphorus concentration, benthic filterer populations, and water temperature. Very low nitrogen concentrations may also hinder macroalgae populations.

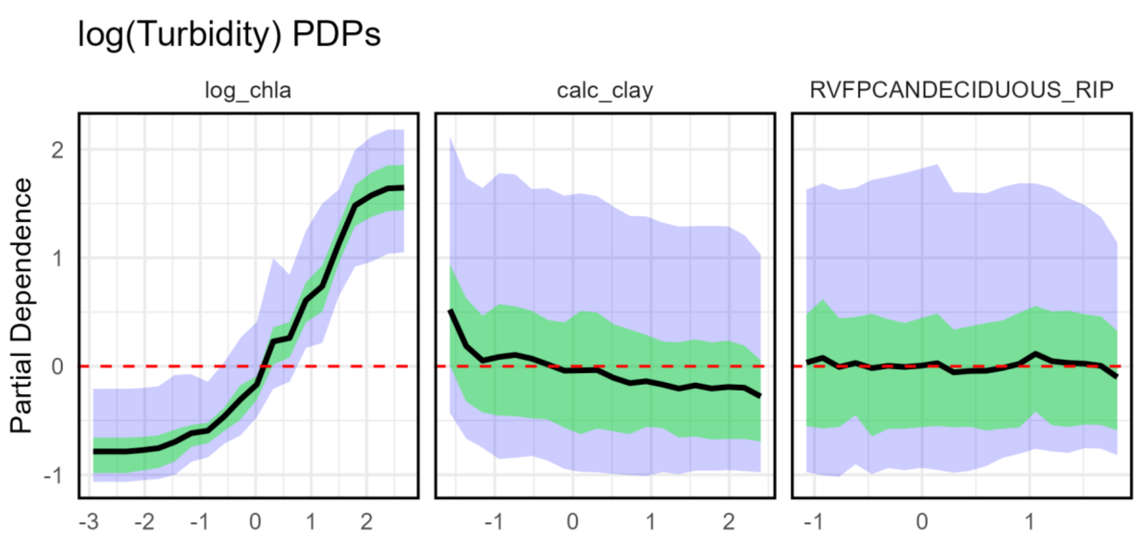


Figure 3a: Partial Dependence Plots for log(Turbidity). Variables are sorted from left-to-right by their feature importance values (most to least important). PD was evaluated during each round of Monte Carlo CV on a common grid of 20 evenly spaced points between the min and max values for

each variable. In the figure, the black line is the mean, green area is the 50% quantile range, and blue area is the 95% quantile range. Values with PD greater than zero (red dashed line) tend to produce above-average response values (and vice versa).

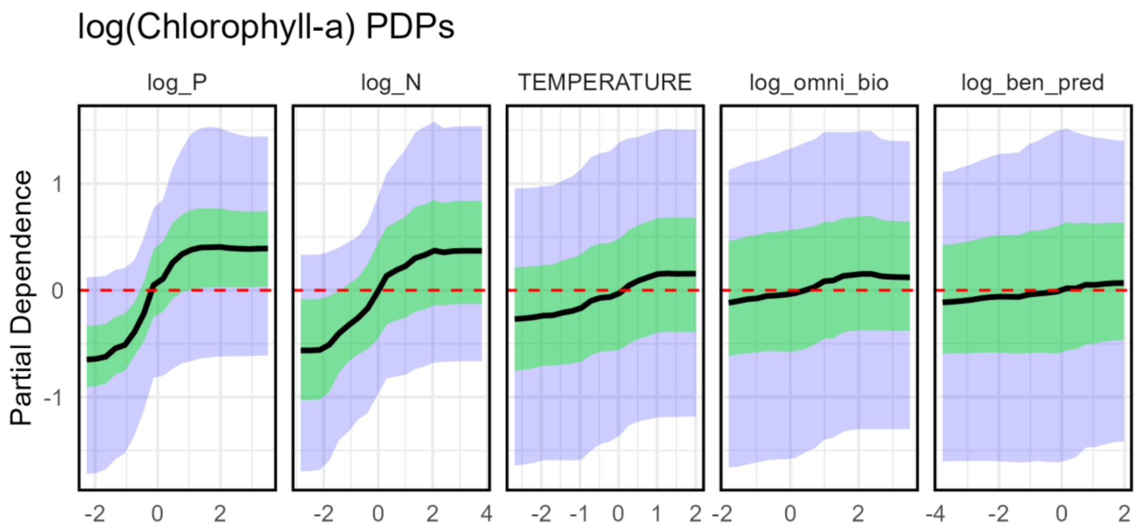


Figure 3b: Partial Dependence Plots for log(Chlorophyll-a). The top 5 features are displayed, ordered by mean feature importance. See Figure 3a caption for more details.

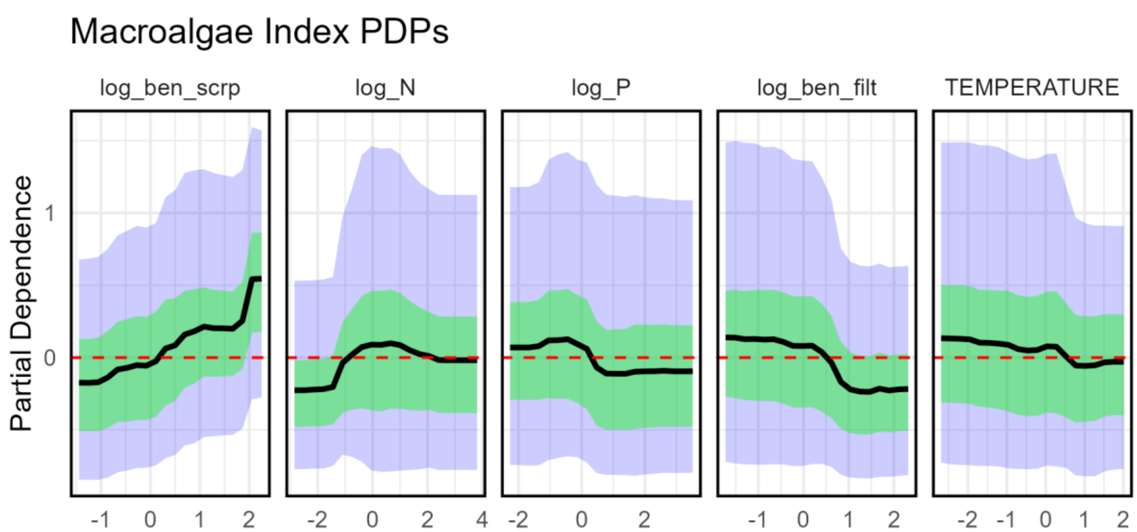


Figure 3c: Partial Dependence Plots for Macroalgae Index. The top 5 features are displayed, ordered by mean feature importance. See Figure 3a caption for more details.

Discussion

Water clarity

The results from this analysis show that lakewater visibility during summer months (May-September) is largely explained by chlorophyll-a concentration (a.k.a. phytoplankton). The proportion of clay substrate has a minimal if not slightly negative impact on turbidity, and deciduous forest cover does not seem to have any impact. This means we can't blame Georgia red clay soils for murky lakes, or brown tannins from deciduous leaf litter. Instead, the recreational experience of U.S. lake-goers is largely influenced by the ecological, geological, and anthropogenic forces that cause phytoplankton growth.

Top-down versus bottom-up control

The second goal of this study was to determine whether bottom-up (i.e. nutrients) or top-down (i.e. food web) mechanisms have more relative influence on phytoplankton growth in U.S. lakes. Analyzing the feature importance and partial dependence plots from the Random Forest models along with the piecewise SEM model coefficients indicate that phosphorus and nitrogen concentrations tend to be by far the most influential factor on phytoplankton populations. Water temperature also plays a small role, and the ecological features have only a marginal impact. These results illustrate that U.S. lakes are dominated by bottom-up mechanisms, and the best way to reduce phytoplankton blooms is by limiting the key nutrients they require.

The models also indicate that macroalgae and phytoplankton might exhibit tradeoffs dependent on phosphorus. Macroalgae and phytoplankton compete for many of the same resources, including nutrient and light access in the water column. Based on the PDPs (Figure 3b & 3c), macroalgae and phytoplankton have an inverse relationship with phosphorus (and water temperature to a lesser extent), where macroalgae tend to be more prevalent in low-phosphorus and colder lakes. This suggests that phytoplankton may outcompete macroalgae for resources when nutrient concentrations and temperatures are high.

There are some potential top-down mechanisms influencing both phytoplankton and macroalgae growth in lake ecosystems. Phytoplankton concentration decreases with herbivorous zooplankton biovolume, but increases with omnivorous and predatory zooplankton biovolume and benthic macroinvertebrate predators that consume the herbivores (Figures 2 & 3b). Macroalgae increases with benthic macroinvertebrate scrapers (e.g. snails, some beetles and other insects), which tend to consume biofilms coating rocks, macroalgae, and aquatic plants (Figures 2 & 3c). Their presence would free up nutrient and light access for macroalgae to take advantage of. On the other hand, macroalgae tend to be negatively impacted by benthic macroinvertebrate filterers (e.g.

mussels, clams, some insect larvae), possibly by consuming macroalgae propagules or some other mechanism. However, these ecological factors are more inconsistent than the nutrient/bottom-up effects and may not be as trustworthy.

Limitations

Ecological food webs are extremely complex, and there may be up to 5 or 6 trophic levels in these lake systems. These interactions are very difficult to model and interpret without high-quality data and subject-matter expertise. Ideally I would have modeled all the specific interactions separately in a more complex DAG (directed acyclic graph) instead of modeling all top-down interactions as direct effects on phytoplankton and macroalgae. These results are good for comparing the magnitudes of different factors on water clarity and phytoplankton growth, but do not produce exact numeric values such as "decreasing nitrogen concentration by 10 mg/L will increase water visibility by 1 foot" since many variables are recorded as percentages or converted to log scale. The piecewise SEM models likely suffer from heteroskedasticity and non-normal residuals even after many feature transformations on count, percentage, and otherwise skewed data, and I did not thoroughly investigate whether the models met these assumptions. Lastly, the bird observation data was very interesting but likely contains many biases that would limit its utility for this analysis – such as more photos being taken of larger, more interesting birds near lakes in highly populated areas. The data was also heavily right-skewed, with more than 75% of lakes having zero observations for each of the focal taxa.

Future work

There is a lot of future work to do with this dataset. I would be very interested to try modeling taxa interactions at a more granular level to further study top-down influences in lake ecosystems. I would also like to partition the lakes by region in a mixed-effects model to search for regional influences on lakewater visibility and phytoplankton growth, and I'd like to do a similar study using only lakes with nonzero bird sightings while also expanding the dataset to include the Cornell Lab of Ornithology's eBird Observation Dataset. It would also be interesting if this NLA dataset was expanded to include winter months so we could explore whether there are seasonal effects on phytoplankton populations or water visibility.

Conclusion

This study demonstrates that U.S. lakewater visibility is primarily influenced by the effect of nutrient concentration on phytoplankton growth, at least during summer months. Excess nitrogen and phosphorus accumulation, typically from anthropogenic sources such as agriculture and livestock operations, seems to be the dominant factor causing phytoplankton blooms. Evidence of top-down regulation is significantly lower in magnitude and more inconsistent, suggesting that these aquatic ecosystems can do relatively little to slow down blooms in eutrophic lakes. In order to create or maintain high quality lake ecosystems it's crucial to create infrastructure, policy, and healthy riparian habitat to reduce the quantity of nutrients reaching these important sources of biodiversity and human recreation.

Lessons Learned

I found this course very useful as a review and extension of key statistical methods introduced in previous courses. It was helpful to see a more rigorous statistical view of these methods, although could be challenging to follow at times. In this report I learned about random forest and piecewise SEM models in greater depth, including partial dependence plots and SHAP values for analyzing per-feature influence on model predictions. I also learned how to aggregate and present these metrics across many rounds of cross validation. Outside of data mining methods, I got to explore multiple very interesting datasets while working on this report and learned a lot about lake ecology. Lastly, I used RSQLite for the first time to organize and merge these large datasets.

Credits

Acknowledgment: The National Lakes Assessment (2022) data were a result of the collective efforts of dedicated field crews, laboratory staff, data management and quality control staff, analysts and many others from EPA, states, tribes, federal agencies, universities, and other organizations. Please contact nars-hq@epa.gov with any questions.

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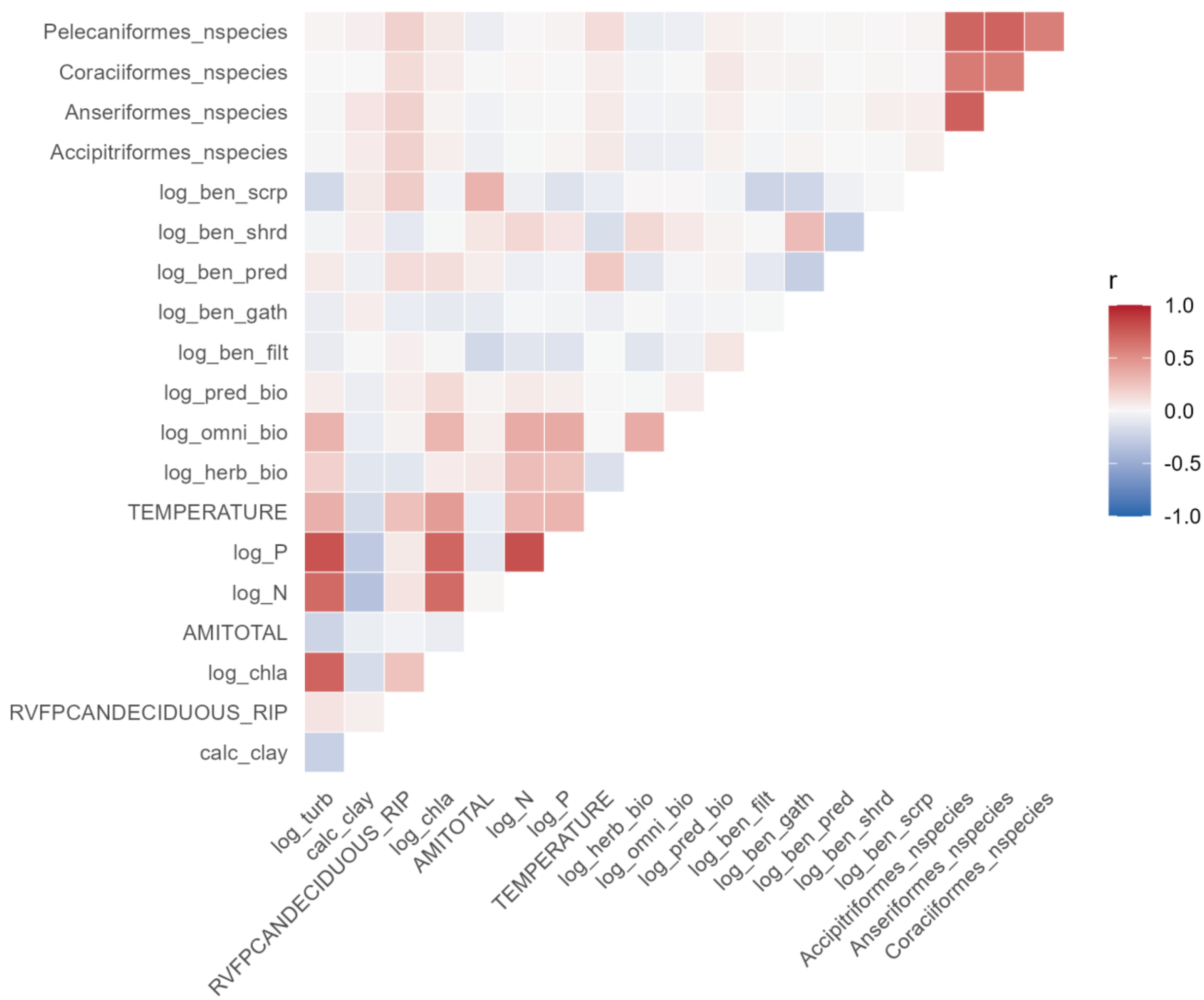
Appendix

Variable	Source	Description	Proxy/Common Term	Transformation
log_turb	nla22_waterchem_wide (TURB_RESULT)	Analyte value: turbidity	Water clarity	log10
calc_clay	nla2022_phabmets_wide_0	fractional cover of sand using bs_sand (mean(wtA, wtB, … wtJ)) fractional cover of silt using bs_silt (mean(wtA, wtB, … wtJ)) Clay is the smallest component of soil and can remain suspended in water for long periods, reducing visibility.	Clay	1 – BSFCSAND - BSFCSILT
RVFPCANDECIDUOUS_RIP	nla2022_phabmets_wide_0	fraction of deciduous canopy presence in the riparian zone using CANOPY (mean(atA, atB… atJ) in the riparian zone) Tannins leach into lakewater from tree leaves and have a brown coloration.	Deciduous forest cover (aka tannins)	
log_chla	nla22_waterchem_wide (CHLA_RESULT)	Analyte value: chlorophyll-a The main type of chlorophyll used by both algae and cyanobacteria (US EPA, 2024). The concentration of chlorophyll-a in lakewater is a consistent indicator of the presence of all types of phytoplankton.	Phytoplankton concentration	log10
AMITOTAL	nla2022_phabmets_wide_0	index of total cover of aquatic macrophytes using am_emergent, am_floating, am_submergent (mean(Aemergent+Afloating+Asubmergent, Bemergent+Bfloating+Bsubmergent, …Jemergent+Jfloating+Jsubmergent))	Macroalgae presence (not likely to be present in the water samples	

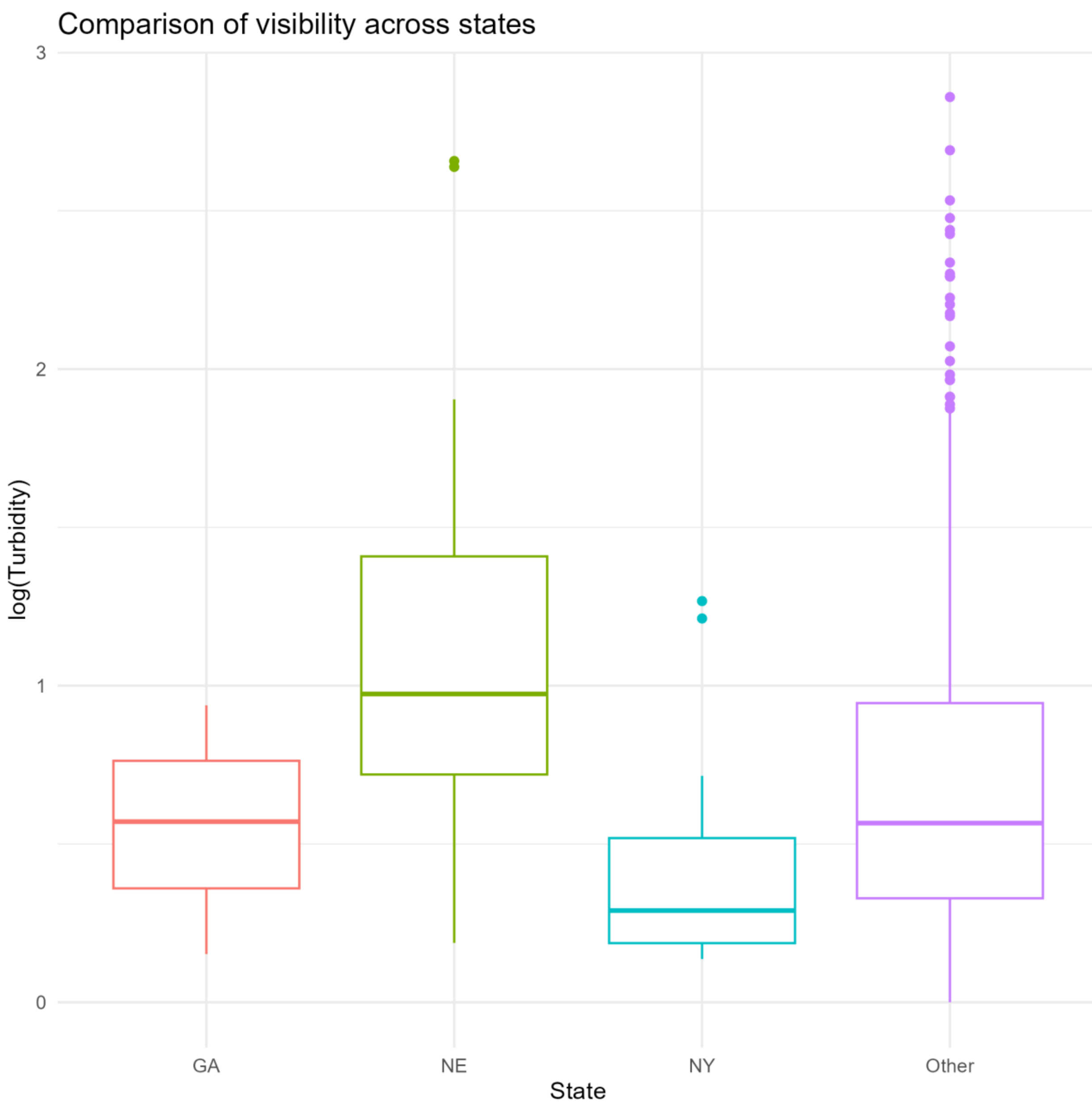
			for turbidity testing)	
log_N	nla22_waterchem_wide (NTL_RESULT)	Result for Total Nitrogen (mg N/L) Total nitrogen concentration in lakewater has been shown in many previous studies to be one of the two key nutrients controlling phytoplankton growth in lakes (along with phosphorus). Nitrogen deposition in lakewater from anthropogenic sources (e.g. agricultural fertilizer and manure runoff) has been linked with increasing phytoplankton growth and harmful algal blooms in U.S. lakes (US EPA, 2024).	Nitrogen	log10
log_P	nla22_waterchem_wide (PTL_RESULT)	Result for Total Phosphorus (ug/L) Total phosphorus concentration is the other key nutrient dictating phytoplankton growth in U.S. lakes (US EPA, 2024). Some research indicates that phosphorus may be even more impactful than nitrogen.	Phosphorus	log10
TEMPERATURE	nla2022_profile_wide	Temperature for index profile	Water temperature	
log_herb_bio	nla22_zooplankton_metrics (HERB_BIO)	Biomass represented by individuals that are herbivores	Herbivorous zooplankton	log10
log_omni_bio	nla22_zooplankton_metrics (OMNI_BIO)	Biomass represented by individuals that are omnivores	Omnivorous zooplankton	log10
log_pred_bio	nla22_zooplankton_metrics (PRED_BIO)	Biomass represented by individuals that are predators	Predatory zooplankton	log10
log_ben_filt	nla22_benthic_metrics (COFIPIND)	% Collecter-filterer individuals\ "Filterers: Percent of the macrobenthos that filter FPOM (fine particulate organic matter) from either the water column or sediment"	Invertebrate filter-feeders	log10
log_ben_gath	nla22_benthic_metrics (COGAPIND)	% Collecter-gatherer individuals Gatherers and filterers: Percent of collector feeders of CPOM and FPOM	Invertebrate gatherers	log10
log_ben_pred	nla22_benthic_metrics (PREDPIND)	% Predators individuals	Invertebrate predators	log10

log_ben_shrd	nla22_benthic_metrics (SHRDPIND)	% Shredders individuals Percent of the macrobenthos that "shreds" leaf litter	Invertebrate shredders (of leaf litter)	log10
log_ben_scrp	nla22_benthic_metrics (SCRPPIND)	% Scrapers individuals Percent of the macrobenthos that scrape or graze upon periphyton	Invertebrate scrapers/grazers	log10
Accipitriformes_nspecies	iNaturalist Research-grade Observations	Number of Accipitriformes species observed within 1 mile of the test site in 2022	Hawks, Eagles, Osprey	
Anseriformes_nspecies	iNaturalist Research-grade Observations	Number of Anseriformes species observed within 1 mile of the test site in 2022	Ducks, Geese, Swans	
Coraciiformes_nspecies	iNaturalist Research-grade Observations	Number of Coraciiformes species observed within 1 mile of the test site in 2022	Kingfishers, etc.	
Pelecaniformes_nspecies	iNaturalist Research-grade Observations	Number of Pelecaniformes species observed within 1 mile of the test site in 2022	Pelicans, Herons, Cormorants	

Appendix 1: Data dictionary.



Appendix 2: Feature correlation coefficients.



Appendix 3: Comparison of water clarity across states. Georgia has average lakewater visibility, Nebraska has the worst visibility, and New York has the best.